

CHECK SUM

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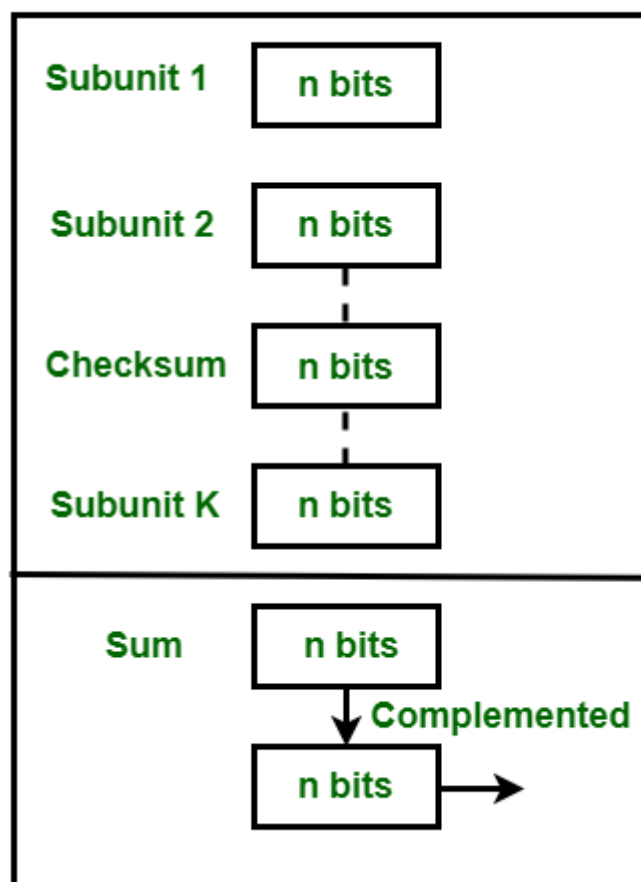
Sender site:

- 1. The message is divided into 16-bit words.**
- 2. The value of the checksum word is set to 0.**
- 3. All words including the checksum are added using one's complement addition.**
- 4. The sum is complemented and becomes the checksum.**
- 5. The checksum is sent with the data.**

Receiver site:

- 1. The message (including checksum) is divided into 16-bit words.**
- 2. All words are added using one's complement addition.**
- 3. The sum is complemented and becomes the new checksum.**
- 4. If the value of checksum is 0, the message is accepted; otherwise, it is rejected.**

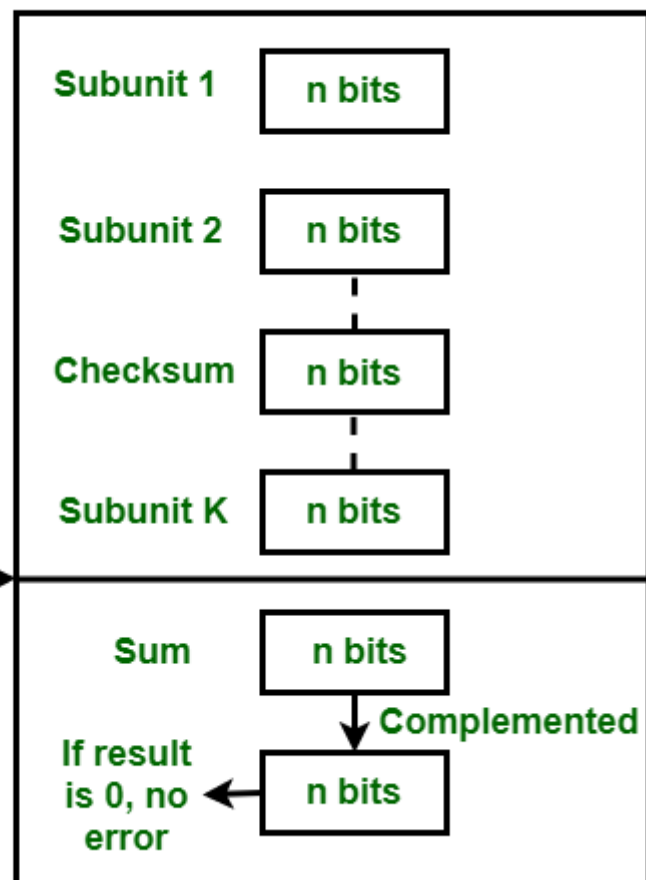
SENDER



Checksum

Data

RECEIVER



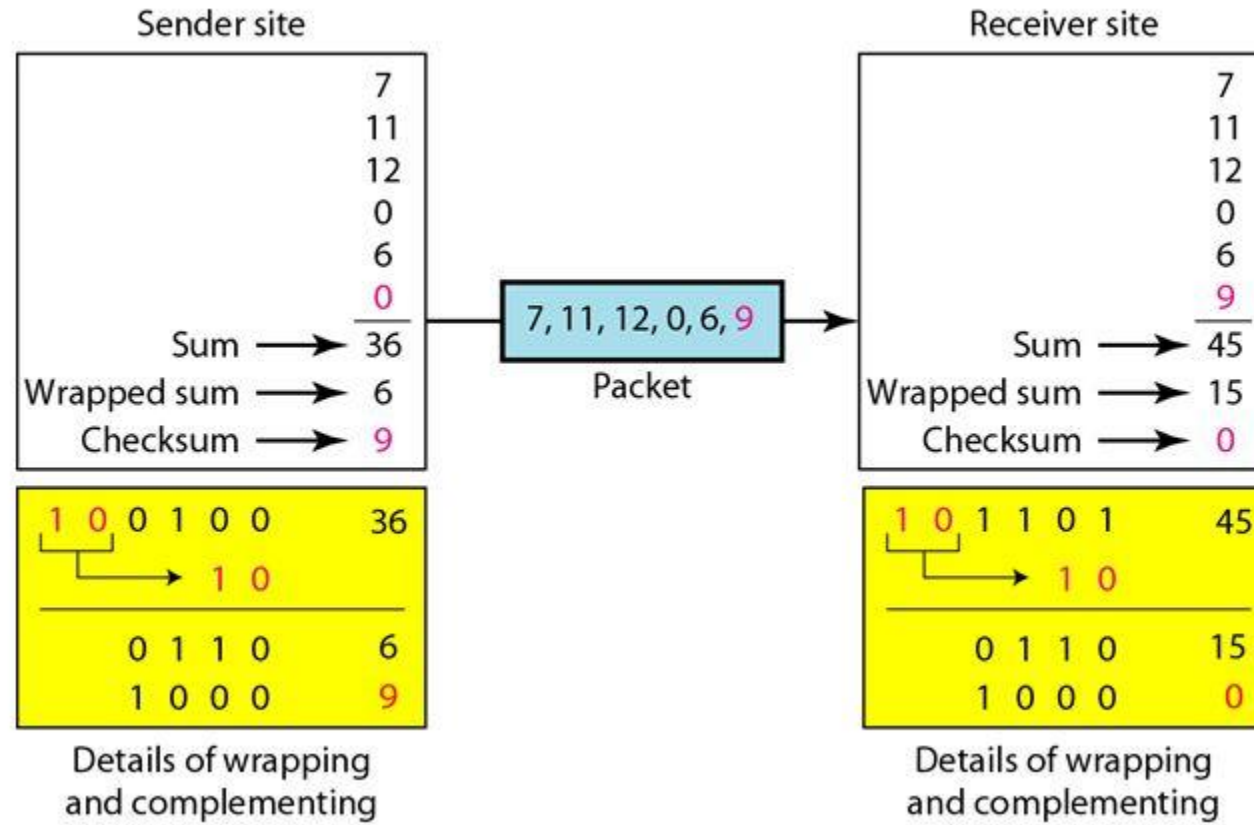
Let us calculate the checksum for a text of 8 characters (“Forouzan”). The text needs to be divided into 2-byte (16-bit) words. We use ASCII (see Appendix A) to change each byte to a 2-digit hexadecimal number. For example, F is represented as 0x46 and o is represented as 0x6F. Figure 10.25 shows how the checksum is calculated at the sender and receiver sites. In part a of the figure, the value of partial sum for the first column is 0x36. We keep the rightmost digit (6) and insert the leftmost digit (3) as the carry in the second column. The process is repeated for each column. Note that if there is any corruption, the checksum recalculated by the receiver is not all 0s. We leave this an exercise.

1	0	1	3	Carries
4	6	6	F	(Fo)
7	2	6	F	(ro)
7	5	7	A	(uz)
6	1	6	E	(an)
0	0	0	0	Checksum (initial)
8	F	C	6	Sum (partial)
8	F	C	7	Sum
7	0	3	8	Checksum (to send)

a. Checksum at the sender site

1	0	1	3	Carries
4	6	6	F	(Fo)
7	2	6	7	(ro)
7	5	7	A	(uz)
6	1	6	E	(an)
7	0	3	8	Checksum (received)
F	F	F	E	Sum (partial)
8	F	E	7	Sum
0	0	0	0	Checksum (new)

a. Checksum at the receiver site



Original Data

10011001	11100010	00100100	10000100
----------	----------	----------	----------

1 2 3 4
k=4, m=8

Sender

1 10011001
2 11100010
① 01111011
1
01111100
3 00100100
10100000
4 10000100
① 00100100
1

Sum: 00100101

Checksum: 11011010

Receiver

1 10011001
2 11100010
① 01111011
1
01111100
3 00100100
10100000
4 10000100
① 00100100
1
00100101
11011010

Sum: 11111111

Complement: 00000000

Conclusion: Accept Data

example: add two 16-bit integers

	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0
	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
wraparound	1	1	0	1	1	1	0	1	1	1	0	1	1	1	0	1
sum	1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0
checksum	0	1	0	0	0	1	0	0	0	1	0	0	0	0	1	1

Note: when adding numbers, a carryout from the most significant bit needs to be added to the result

Solution to example 1

1. Assume a packet is made only of four 16-bit words $(3456)_{16}$, $(ABCC)_{16}$, $(02BC)_{16}$, and $(EEEE)_{16}$. Manually simulate the algorithm in Figure 10.17 to find the checksum.

a. Checksum at the sender site

1	2	2	2	Carries
3	4	5	6	
A	B	C	C	
0	2	B	C	
E	E	E	E	
0	0	0	0	Checksum (initial)
D	1	C	C	Sum (partial)
D	1	C	D	Sum
2	E	3	2	Checksum (to send)

b. Checksum at the receiver site (no error)

1	2	2	2	Carries
3	4	5	6	A B C D E F
A	B	C	C	10 11 12 13 14 15
0	2	B	C	
E	E	E	E	
2	E	3	2	Checksum (received)
F	F	F	E	Sum (partial)
F	F	F	F	Sum
0	0	0	0	Checksum

```

10101001
00111001
11100010 ← sum
00011101 ← checksum

```

Pattern sent is 10101001 00111001 00011101.

The receiver does the following operations :

```

10101001
00111001
00011101
11111111
00000000 ← result

```

So in this case there is no error and the data is accepted.

In case of error suppose the received data is 10101111 11111001 00011101. Then the receiver will compute like this :

```

10101111
11111001
00011101
111000101
  1
11000110
00111001

```

1. Assume a packet is made only of four 16-bit words $(3456)_{16}$, $(ABCC)_{16}$, $(02BC)_{16}$, and $(EEEE)_{16}$. Manually simulate the algorithm in Figure 10.17 to find the checksum.

Checksum

Sending:

1. Arrange data in 16-bit words
2. Put zero in checksum position, add
3. Add any carryover back to get 16 bits
4. Negate (complement) to get sum

0001
f203
f4f5
f6f7

Checksum (3)

Sending:

1. Arrange data in 16-bit words
2. Put zero in checksum position, add
3. Add any carryover back to get 16 bits
4. Negate (complement) to get sum

```
0001
f203
f4f5
f6f7
+ (0000)
-----
2ddf0
  ↓
ddf0
+    2
-----
ddf2
  ↓
220d
```

Internet Checksum (4)

Receiving:

1. Arrange data in 16-bit words

2. Checksum will be non-zero, add

```
0001
f203
f4f5
f6f7
+ 220d
-----
```

3. Add any carryover back to get 16 bits

4. Negate the result and check it is 0

Internet Checksum (5)

Receiving:

1. Arrange data in 16-bit words
2. Checksum will be non-zero, add
3. Add any carryover back to get 16 bits
4. Negate the result and check it is 0

```
0001
f203
f4f5
f6f7
+ 220d
-----
2fffd
  ↓
  fffd
+    2
-----
ffff
  ↓
0000
```

Checksum Problems

1. 466f, 726f, 757A, 616E

2. 0001

- f203
- f4f5
- f6f7

757A ✓
616E ✓
0000 ← checksum
8FC6 ← Sum
1

726F ✓
757A ✓
616E ✓
0000 ← checksum (initial)
8FC6 ← Sum
1
8FC7 (Sum)
7038 (checksum to SEND)

① ^{0 1 3}
466F-

726F-

757A-

616E-

7038 ← Checksum

FFFFE

→ + 1

$$\begin{array}{r} \textcircled{3} \\ 16 \overline{) 62} \\ \underline{48} \\ 14 \end{array}$$

$$\begin{array}{r} 1 \\ 16 \overline{) 31} \\ \underline{16} \\ 15 \end{array}$$

$$\begin{array}{r} 1 \\ 16 \overline{) 31} \\ \underline{16} \\ 15 \end{array}$$

Checksum Problems

- 00FE, C523,FDA1,D68A,AF02

00FE C523 FDA1 D68A AF02

$$\begin{array}{r} 00\overset{!}{F}\overset{!}{E} \\ + C523E \end{array}$$

'C 621

+ F D A I

10302

$$L \rightarrow +1$$

C3C3

+ D68A

19A4D

$$\hookrightarrow +1$$

19A'4E

+ AF02

14950

$$L \rightarrow +$$

4951

ones comp

B6AE

00FE	C523	FDA1	D68A	AF02
00FE				
+ C523	←			

C621				
+ FDA1	←			

1C3C2				
↳ +1				

C3C3				
+ D68A	←			

19A4D				
↳ +1				

19A4E				
+ AF02	←			

14950				
↳ +1				

4951				
		ones comp		
		→		
			(B6AE)	←

$$\text{BDP} = B * D$$

- Where BDP is the bandwidth-delay product
- B is the bandwidth or data link capacity (bits per second)
- D is the round-trip delay (seconds)

BANDWIDTH-DELAY PRODUCT – SOLVED EXAMPLE

Consider that the link capacity of a channel is 512 Kbps and round – trip delay time is 1000ms.

Solution:

$$\begin{aligned}\text{The bandwidth delay product} &= 512 \text{ Kbps} \times 1000 \text{ ms} \\ &= 512 \times 1000 \text{ bits/sec} \times 1000 \times 10^{-3} \text{ sec} \\ &= 512,000 \text{ bits} \\ &= 64,000 \text{ bytes} \\ &= 62.5 \text{ KB}\end{aligned}$$

b)

Round trip time, $RTT = 20$ ms Bandwidth, $B = 10$ Mbps Packet size, $P = 2000$ b Bandwidth-delay product, $BP = ?$

Calculating BP :

$$BP = B \cdot RTT = 10 \cdot 10^6 \cdot 20 \cdot 10^{-3} = 200000 \text{ b}$$

BP in packets:

$$BP = \frac{BP}{P} = 100$$

(2) Bandwidth-delay calculation for the network Ethernet:

Convert the bandwidth value into bits.

$$\begin{aligned}\text{bandwidth value} &= 10\text{Mbps} \times 1,000\text{kilobits} \times 1,000\text{bits} \\ &= 10,000,000 \text{ bps}\end{aligned}$$

Substitute the Ethernet bandwidth value as 10 Mbps and bandwidth delay as 0.1 sec in Equation (1).

$$\begin{aligned}\text{Bandwidth delay} &= 10,000,000 \text{ bits} \times 0.1 \text{ sec} \\ &= 1,000,000 \text{ bits} \\ &= 1,000,000/1000 \text{ kilobits} \\ &= 1,000 \text{ kilobits} \\ &= 1000/8 \text{ bytes} \\ &= 125 \text{ kilobytes}\end{aligned}$$

Thus, the bandwidth delay for the Ethernet network is **125 kilobytes**.

(1) Bandwidth-delay calculation for the network T1:

Convert the bandwidth value into bits.

$$\begin{aligned}\text{bandwidth value} &= 1.5 \text{ Mbps} \times 1000 \text{ kilobits} \times 1000 \text{ bits} \\ &= 1,500,000 \text{ bps}\end{aligned}$$

Substitute the T1 bandwidth value as 1.5 Mbps and bandwidth delay as 0.1 sec in equation formula (1).

$$\begin{aligned}\text{Bandwidth delay} &= 1,500,000 \text{ bits} \times 0.1 \text{ sec} \\ &= 150,000 \text{ bits} \\ &= 150,000/1000 \text{ kilobits} \\ &= 150 \text{ kilobits} \\ &= 150/8 \text{ bytes} \\ &= 18.75 \text{ kilobytes}\end{aligned}$$

Practice Problems Contd

- A network has a bandwidth of 10 Mbps and a round-trip delay of 100 ms. What is the Bandwidth Delay Product?
- $BDP = \text{Bandwidth} \times \text{Round-trip Delay}$
 $BDP = 10\text{Mbps} \times 100\text{ ms}$
 $BDP = 1,25000\text{ bytes}$

Practice Problems Contd

- A network has a bandwidth of 100 Mbps and a round-trip delay of 10 ms. What is the Bandwidth Delay Product?
- BDP = Bandwidth x Round-trip Delay
BDP = 100 Mbps x 10 ms
BDP = 12,500 bytes

Practice Problems Contd

- A network has a bandwidth of 1 Gbps and a round-trip delay of 1 ms. What is the Bandwidth Delay Product?
- $BDP = \text{Bandwidth} \times \text{Round-trip Delay}$
 $BDP = 1 \text{ Gbps} \times 1 \text{ ms}$
 $BDP = 125,000 \text{ bytes}$
- As we can see from these examples, the Bandwidth Delay Product is dependent on both the bandwidth and round-trip delay of the network. The higher the bandwidth and lower the delay, the larger the Bandwidth Delay Product.